

Shivam Sourav

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PROFESSIONAL EXPERIENCE

Associate Software Engineer

Aug 2024 – Present

Nomura Research Institute and Financial Technology

On site - Kolkata

- Automated release deployments via **Jenkins CI/CD**, reducing manual effort by **70%** with pipelines.
- Designed **Python**-based data generation utility using **YAML** schemas and configurable column definitions, supporting **OracleDB**, and enabling **60% faster** high-volume data creation with advanced formatting and mapping logic
- Implemented **LLM-based ATS** with **LlamaIndex**, **Cohere**, and **Pinecone**, enabling **90%+ accurate** resume matching, real-time scoring, and **AI-driven recommendations** via **Next.js** and **Flask**

Data Scientist

Nov 2023 – Jun 2024

Omdena

Remote

- Designed **time-series models** using **XGBoost** for COVID-19 forecasting and **LSTM** for flood prediction, achieving **92–95% accuracy** via feature engineering and **GridSearchCV** tuning
- Spearheaded a cross-functional team of 10 in **data preprocessing** and **model deployment, streamlining** processes that enhanced collaboration among 40 colleagues by improving workflow efficiency by 50% through targeted training sessions on **Pandas** and **Scikit-learn usage**.

EDUCATION

Sikkim Manipal Institute of Technology

May 2025

B.Tech in Artificial Intelligence and Data Science

CGPA: 9.7

UNIVERSITY PROJECTS

Nepali LLM - Tuned Language Model

- Trained a **SentencePiece tokenizer** on Nepali corpus, reducing token count by **80%** and fine-tuned **Gemma-2B** via **LoRA**, achieving **25% higher accuracy** on NLU tasks
- Deployed **Nepali chatbot** with **Streamlit**, increasing user engagement by **30%**

Vitalis - Clinical AI Dashboard

- Engineered **Flask API** implementing **clinical algorithms** (ASCVD/CKD-EPI) processing **50+ biomarkers/sec**, achieving **92% concordance** with clinician judgments through **NumPy-optimized** calculations
- Deployed **Qwen2.5 3B** via **Ollama** (4-bit GGUF) to produce **JSON** outputs including **NLP summaries** (95% Score), **trend alerts**, and **clinical recommendations**, with optimal inference speed.
- Architected **React** dashboard with **Chart.js/D3.js** visualizations (risk heatmaps, trend graphs) connected to **Flask API**, displaying real-time clinical data with 200ms refresh intervals

Brainy-Buddy - RAG Document Chatbot

- Formulated a **RAG-based Q&A system** using **LlamaIndex**, **Cohere embeddings**, and **Pinecone**, with **Groq-powered LLM inference**, achieving **85% top-3 retrieval precision** and real-time semantic justifications via **Streamlit**

News Headline Generator

- Devised fine-tuning for **Mistral-7B** on news summarization using **QLoRA (rank=64)** and **Unsloth**, achieving **80% headline relevance** with **50% faster training**

ADDITIONAL

- **Languages & Tools:** Python, Java, R, SQL, Git, GitHub, Jenkins, Puppet, Apache Tomcat, ActiveMQ
- **Frameworks & Libraries:** TensorFlow, PyTorch, Keras, Scikit-Learn, XGBoost, Prophet, Hugging Face, NLTK, LlamaIndex, Flask, Streamlit, Gradio
- **Data & Visualization:** Pandas, NumPy, MongoDB, OracleDB, Power BI, Tableau, Looker, Plotly, Matplotlib, Seaborn, Chart.js
- **Concepts & Soft Skills:** Machine Learning, Deep Learning, NLP, Computer Vision, Time Series, Statistics, Big Data, Team Collaboration, Leadership